Code K-Means - Minimum Nécessaire

AmineGR03

5 février 2026

Code Python

Code Python

```
# Imports
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
from sklearn.datasets import make_blobs

# 1. Chargement du dataset
X, y_true = make_blobs(n_samples=300, centers=4, n_features=2,
    random_state=42)

# 2. Normalisation
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Division train/test
X_train, X_test, _, _ = train_test_split(X_scaled, y_true,
    test_size=0.2, random_state=42)

# 4. K-Means
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
kmeans.fit(X_train)

# 5. Predictions
y_train_pred = kmeans.predict(X_train)
y_test_pred = kmeans.predict(X_test)

# 6. Evaluation
silhouette_train = silhouette_score(X_train, y_train_pred)
silhouette_test = silhouette_score(X_test, y_test_pred)
print(f"Score silhouette (train): {silhouette_train:.4f}")
print(f"Score silhouette (test): {silhouette_test:.4f}")

# 7. Visualisation
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train_pred, cmap='
    viridis', alpha=0.6)
```

```
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_
    [:, 1],
            c='red', marker='x', s=200, linewidths=3, label='
                Centroides')
plt.title(f'Clustering Train (Silhouette: {silhouette_train:.3f})')
plt.legend()

plt.subplot(1, 2, 2)
plt.scatter(X_test[:, 0], X_test[:, 1], c=y_test_pred, cmap='
    viridis', alpha=0.6)
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_
    [:, 1],
            c='red', marker='x', s=200, linewidths=3, label='
                Centroides')
plt.title(f'Clustering Test (Silhouette: {silhouette_test:.3f})')
plt.legend()

plt.tight_layout()
plt.show()
```